

HOSPITAL BED OCCUPANCY MISSING-DATA REPAIR

Advanced Data Science Project

Project ID: Mtech-DS26004

Student: Ahmed Masood

Version 1.0 | Python 3.10+ | Tkinter | scikit-learn

1. Project Information

This portfolio project delivers a reproducible data-repair benchmark, leakage-aware critical-occupancy classifier, interactive administrator desktop application, and governed export workflow.

2. Executive Summary

The synthetic multi-hospital census contains 5,400 daily ward records. Targeted fields contain 4.54% missing cells overall, generated through MCAR, MAR, MNAR, sequential, departmental, weekend, holiday, and block mechanisms. 9.8% of observed records meet the project critical threshold.

3. Problem Statement and Objectives

Delayed entry, sensor failure, incomplete discharge records, synchronization errors, and human mistakes can distort capacity decisions. The objective is to measure repair error and bias through repeated artificial masking, select a defensible method per numerical column, and predict critical occupancy without target leakage.

4. Stakeholders

Hospital administrators need capacity alerts; ward managers need operational detail; data stewards need missingness diagnostics; analysts need reproducible comparisons; clinicians require transparent limitations and human oversight.

5. Dataset Description

Five hospitals, six departments, 180 daily census dates, operational capacity, ICU, flow, staffing, outcomes, climate, calendar, and outbreak fields. Constraints ensure occupied beds do not exceed total beds, ICU occupancy does not exceed ICU capacity, and bed balances remain non-negative.

6. Data Quality Findings

The reusable validator checks identifiers, dates, duplicates, negative values, categorical domains, bed and ICU limits, staffing consistency, balance equations, occupancy bounds, and robust extreme outliers. Missing values are preserved until benchmarked; impossible values are conservatively converted or bounded.

7. Missing-Data Patterns

The generator records known mechanisms. Chi-square associations between missing indicators and department, hospital, season, weekend, and holiday provide MAR-like evidence. Mortality and length of stay include MNAR-designed masking; observed data alone cannot prove or fully recover MNAR values.

8. Methodology

Known values are repeatedly hidden using deterministic seeds. Each method imputes the masked values, which are then compared with their ground truth. Per-column scoring combines RMSE, MAE, absolute bias, KS distance, variance distortion, correlation preservation, and runtime.

9. Statistical Analysis

Descriptive statistics characterize central tendency and shape. Shapiro-Wilk tests normality on a bounded sample; Welch t-test and Mann-Whitney compare weekend occupancy; ANOVA and Kruskal-Wallis compare hospitals and departments; chi-square tests categorical association; KS compares distributions; confidence intervals quantify uncertainty.

10. Imputation Methods

The engine implements mean, median, mode, forward fill, backward fill, linear, time, and polynomial interpolation, KNN, iterative chained equations, department mean, hospital median, seasonal median, rolling median, and a hybrid data-aware strategy.

11. Benchmark Results and Best Method per Column

Rank 1 is selected independently for each column. Lower error and bias increase the score; stronger distribution and correlation preservation increase the score; runtime is a smaller operational penalty.

Column	Method	MAE	RMSE	Bias	Score
ICU_beds	knn	0.326	0.621	0.018	79.3
ICU_occupied	iterative	0.630	0.881	-0.047	76.8
available_beds	iterative	0.000	0.000	-0.000	80.2
average_length_of_stay	department_mean	1.805	2.361	-0.016	66.5
daily_admissions	iterative	1.572	1.896	0.015	74.8
daily_discharge	iterative	1.449	1.862	0.044	81.0
disease_outbreak_flag	rolling_median	0.004	0.047	-0.003	85.8
doctors_on_duty	iterative	0.537	0.670	0.004	76.1
emergency_beds	rolling_median	0.664	1.181	-0.005	85.3
emergency_cases	rolling_median	2.684	3.445	0.024	67.0
holiday_flag	median	0.031	0.173	-0.031	82.7
infection_rate	forward_fill	2.940	3.900	-0.171	85.4
mortality_rate	seasonal	1.574	2.192	-0.457	64.3
nurses_on_duty	iterative	0.860	1.109	0.003	89.1
occupancy_rate	iterative	2.069	2.887	-0.038	79.2
occupied_beds	iterative	0.000	0.000	0.000	88.5
patient_turnover_rate	iterative	3.925	6.059	0.206	78.5
reserved_beds	iterative	0.000	0.000	0.000	75.7

12. Bias Analysis

A method can achieve low average error while systematically over- or under-estimating capacity. Mean bias, percentage bias, variance and standard-deviation distortion, KS statistic, Wasserstein distance, and correlation preservation are therefore reported with point-error metrics.

13. Feature Engineering

Operational ratios, pressure indices, staffing gaps, shortage and risk flags are supplemented by lags and rolling means. All temporal features shift before rolling or use expanding past-only aggregates to prevent look-ahead leakage.

14. Machine Learning Methodology

Seven classifier families are supported through scikit-learn pipelines and a column transformer. ID, date, occupied beds, available beds, occupancy rate, and threshold-derived flags are excluded from predictors. Stratified train, validation, and test partitions are fixed by random seed.

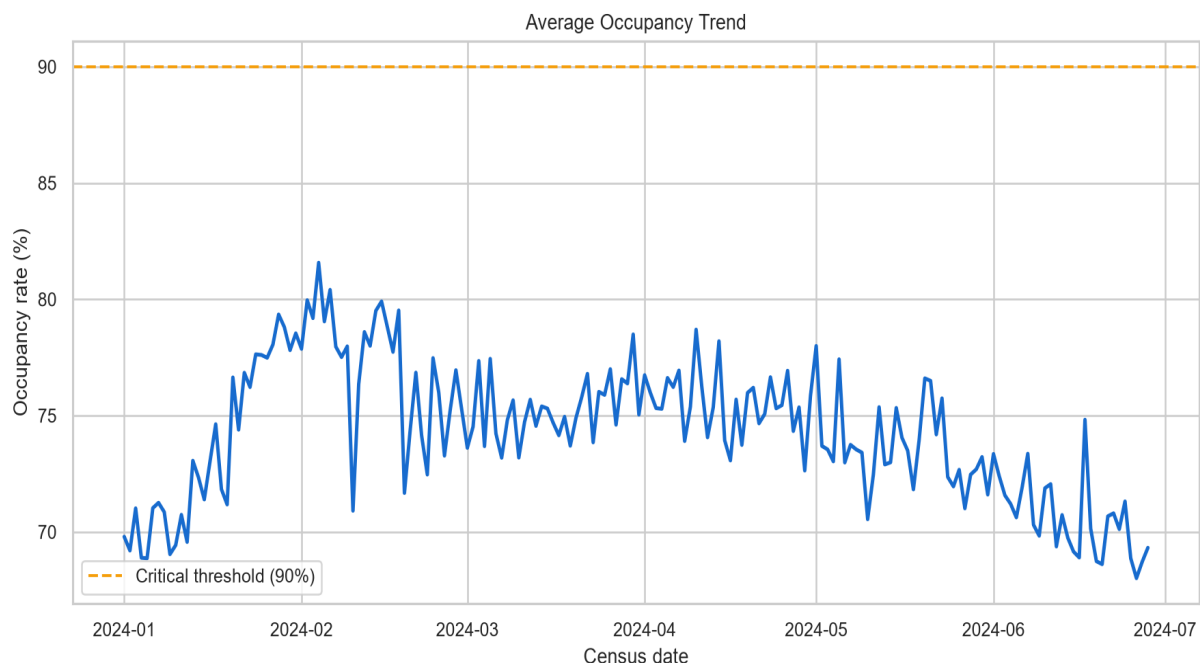
15. Model Evaluation

Accuracy, precision, recall, specificity, F1, ROC-AUC, PR-AUC, balanced accuracy, MCC, log loss, Brier score, calibration, confusion matrix, ROC, and precision-recall curves are saved. Selection weights recall at 60% and ROC-AUC at 40%, reflecting the cost of missed critical events while monitoring false alarms.

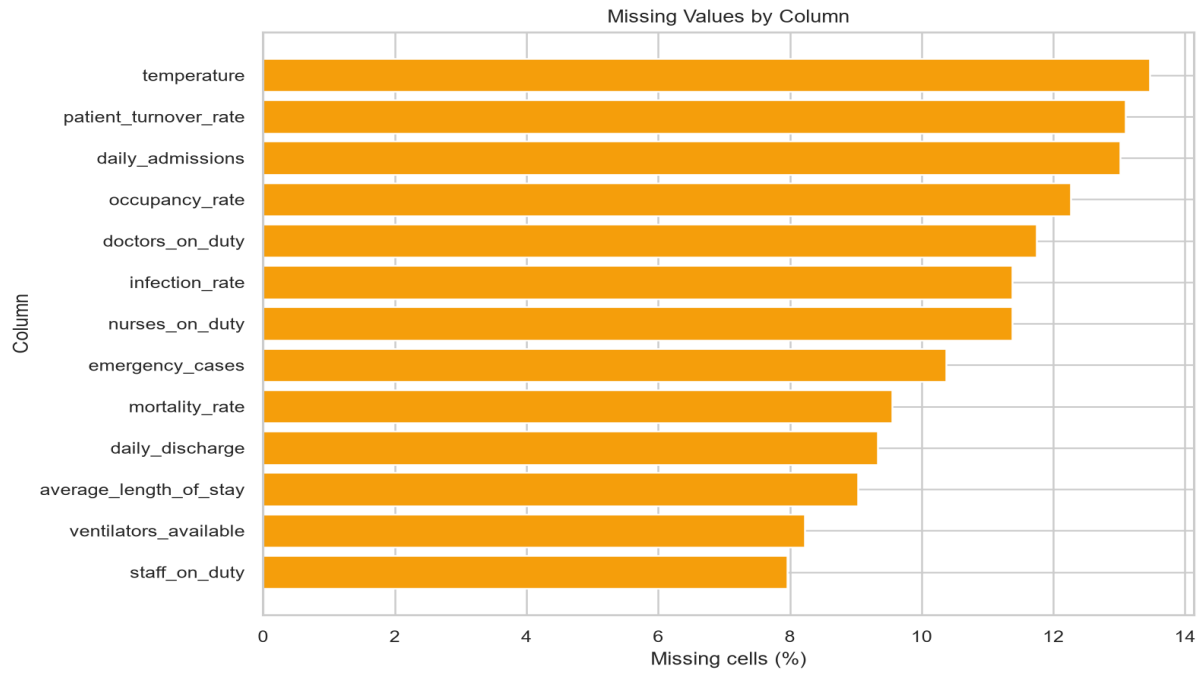
16. Ethical Considerations

Outputs support operational review and must not be treated as clinical decisions. Protected attributes are absent; synthetic records contain no real patient identities. Human review is required for MNAR repair, long gaps, and high-risk alerts. Local governance should set thresholds and monitor subgroup error.

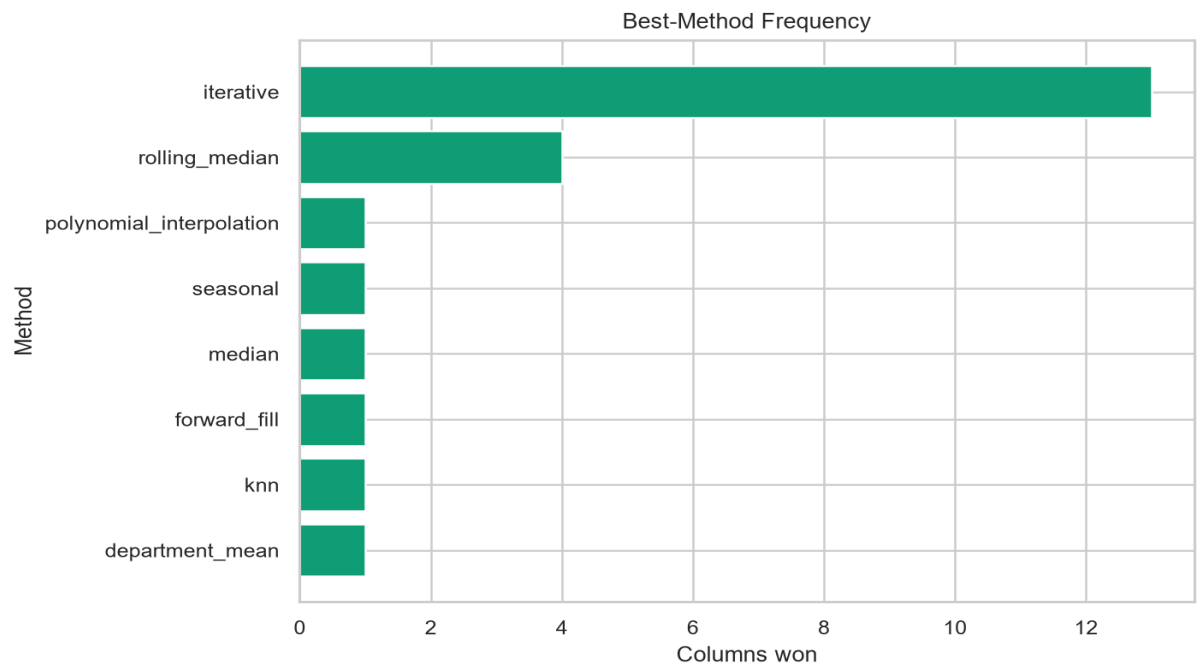
Average occupancy over time with the project threshold.



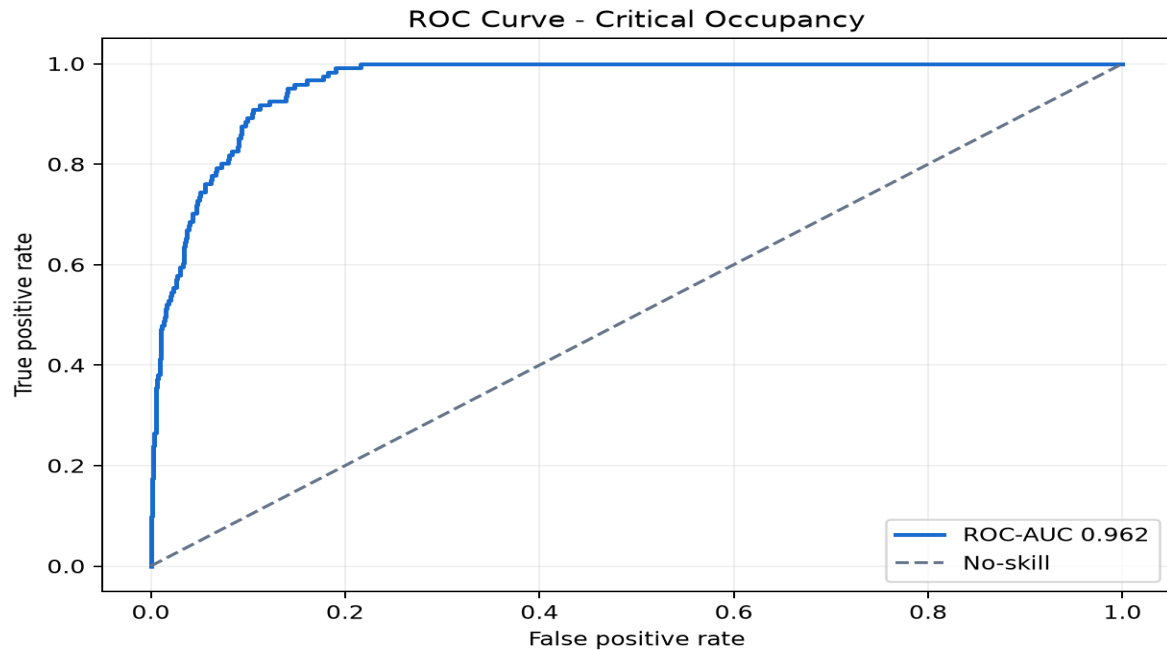
Targeted missingness percentages by field.



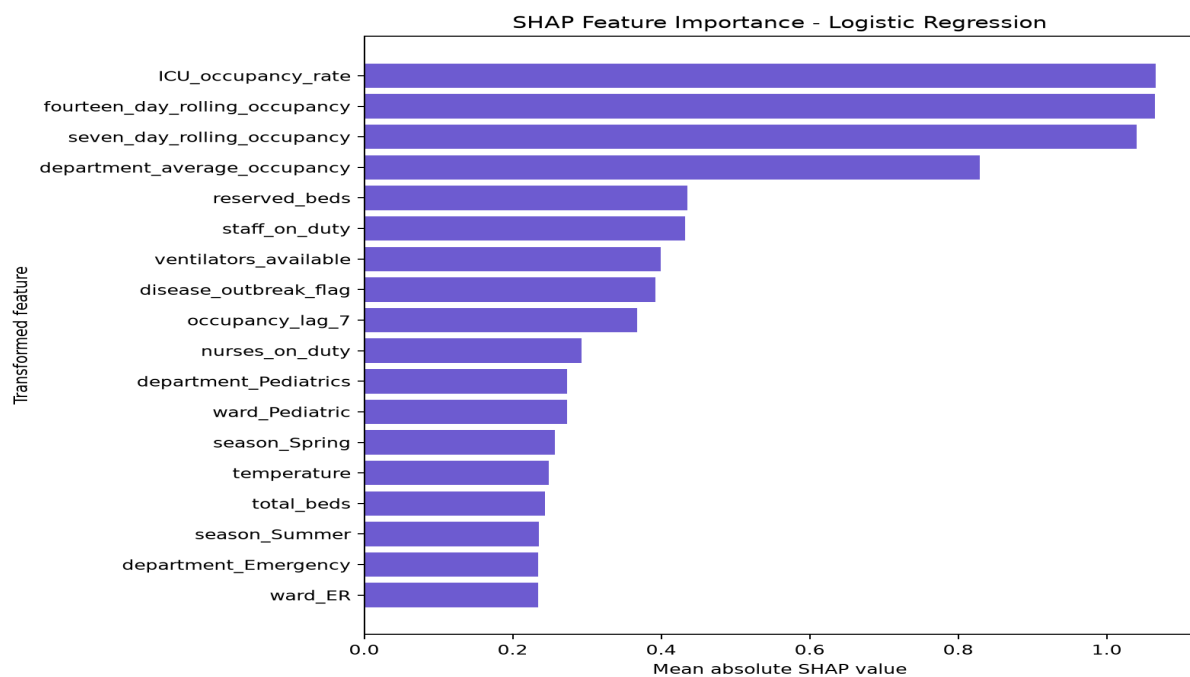
Frequency of per-column benchmark winners.



Out-of-sample discrimination of the selected model.



Mean absolute SHAP importance for the selected model.



17. Business Recommendations

- Use department-level repair for staffing fields when repeated-mask validation confirms lower error.
- Use temporal interpolation only for short gaps; route long sequential gaps and MNAR-designed fields to manual review.
- Avoid mean imputation for skewed length-of-stay, mortality, and infection distributions.
- Trigger capacity escalation at predicted critical risk $\geq 42\%$, then review ICU, admissions, emergency load, and staffing.

- Add mandatory-field validation, synchronization monitoring, and weekend staffing-data ownership at data entry.

18. Assumptions

- The synthetic relationships approximate Pakistani multi-hospital operations but are not clinical ground truth.
- The 90% critical threshold is a project convention and requires hospital-specific governance.

19. Limitations

- Synthetic data may not represent every hospital.
- MNAR values cannot be reliably recovered from observed data alone.
- Imputation cannot replace proper collection.
- KNN and iterative methods can be computationally expensive.
- Interpolation can fail across long gaps.
- Model performance depends on data quality.
- Critical thresholds vary by hospital, ward, and local policy.

20. Future Improvements

Validate on governed real-world hospital data; add temporal cross-validation, drift monitoring, uncertainty intervals, cost-sensitive thresholds, role-based access, database-backed audit trails, and hospital-specific calibration.

21. Conclusion

The project demonstrates that missing-data repair should be selected per field through ground-truth masking and bias-aware metrics, not by a single default. The integrated benchmark, model, GUI, tests, and report form a reproducible decision-support prototype.

22. References

scikit-learn documentation; pandas documentation; SciPy statistical reference; Little and Rubin, Statistical Analysis with Missing Data; ReportLab User Guide; project source code and generated artifacts.